

bigmart-sales-data

July 2, 2024

```
[1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
import seaborn as sns
sns.set(color_codes=True)
```

```
[2]: #load the Dataset
df=pd.read_csv(r"C:\Users\dutta\Downloads\Test.csv")
```

```
[3]: df
```

```
[3]:      Item_Identifier  Item_Weight  Item_Fat_Content  Item_Visibility  \
0          FDW58         20.750         Low Fat         0.007565
1          FDW14          8.300             reg         0.038428
2          NCN55         14.600         Low Fat         0.099575
3          FDQ58          7.315         Low Fat         0.015388
4          FDY38          NaN         Regular         0.118599
...          ...          ...          ...          ...
5676         FDB58         10.500         Regular         0.013496
5677         FDD47          7.600         Regular         0.142991
5678         NC017         10.000         Low Fat         0.073529
5679         FDJ26         15.300         Regular         0.000000
5680         FDU37          9.500         Regular         0.104720
```

```
      Item_Type  Item_MRP  Outlet_Identifier  \
0      Snack Foods  107.8622          OUT049
1          Dairy   87.3198          OUT017
2          Others  241.7538          OUT010
3      Snack Foods  155.0340          OUT017
4          Dairy  234.2300          OUT027
...          ...          ...          ...
5676      Snack Foods  141.3154          OUT046
5677      Starchy Foods  169.1448          OUT018
5678  Health and Hygiene  118.7440          OUT045
5679          Canned  214.6218          OUT017
5680          Canned   79.7960          OUT045
```

	Outlet_Establishment_Year	Outlet_Size	Outlet_Location_Type	\
0	1999	Medium	Tier 1	
1	2007	NaN	Tier 2	
2	1998	NaN	Tier 3	
3	2007	NaN	Tier 2	
4	1985	Medium	Tier 3	
...	...	...	...	
5676	1997	Small	Tier 1	
5677	2009	Medium	Tier 3	
5678	2002	NaN	Tier 2	
5679	2007	NaN	Tier 2	
5680	2002	NaN	Tier 2	

	Outlet_Type
0	Supermarket Type1
1	Supermarket Type1
2	Grocery Store
3	Supermarket Type1
4	Supermarket Type3
...	...
5676	Supermarket Type1
5677	Supermarket Type2
5678	Supermarket Type1
5679	Supermarket Type1
5680	Supermarket Type1

[5681 rows x 11 columns]

```
[4]: # the first 5 rows of the Dataset
df.head()
```

	Item_Identifier	Item_Weight	Item_Fat_Content	Item_Visibility	Item_Type	\
0	FDW58	20.750	Low Fat	0.007565	Snack Foods	
1	FDW14	8.300	reg	0.038428	Dairy	
2	NCN55	14.600	Low Fat	0.099575	Others	
3	FDQ58	7.315	Low Fat	0.015388	Snack Foods	
4	FDY38	NaN	Regular	0.118599	Dairy	

	Item_MRP	Outlet_Identifier	Outlet_Establishment_Year	Outlet_Size	\
0	107.8622	OUT049	1999	Medium	
1	87.3198	OUT017	2007	NaN	
2	241.7538	OUT010	1998	NaN	
3	155.0340	OUT017	2007	NaN	
4	234.2300	OUT027	1985	Medium	

Outlet_Location_Type	Outlet_Type
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0          Tier 1 Supermarket Type1
1          Tier 2 Supermarket Type1
2          Tier 3      Grocery Store
3          Tier 2 Supermarket Type1
4          Tier 3 Supermarket Type3

```

```

[5]: # the first 5 rows of the Dataset
df.tail()

```

```

[5]:      Item_Identifier  Item_Weight  Item_Fat_Content  Item_Visibility  \
5676          FDB58          10.5          Regular          0.013496
5677          FDD47           7.6          Regular          0.142991
5678          NC017          10.0          Low Fat          0.073529
5679          FDJ26          15.3          Regular          0.000000
5680          FDU37           9.5          Regular          0.104720

```

```

      Item_Type  Item_MRP  Outlet_Identifier  \
5676    Snack Foods  141.3154          OUT046
5677    Starchy Foods  169.1448          OUT018
5678  Health and Hygiene  118.7440          OUT045
5679          Canned  214.6218          OUT017
5680          Canned   79.7960          OUT045

```

```

      Outlet_Establishment_Year  Outlet_Size  Outlet_Location_Type  \
5676                1997          Small          Tier 1
5677                2009          Medium          Tier 3
5678                2002           NaN          Tier 2
5679                2007           NaN          Tier 2
5680                2002           NaN          Tier 2

```

```

      Outlet_Type
5676  Supermarket Type1
5677  Supermarket Type2
5678  Supermarket Type1
5679  Supermarket Type1
5680  Supermarket Type1

```

```

[6]: df.info()

```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5681 entries, 0 to 5680
Data columns (total 11 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Item_Identifier        5681 non-null  object
1   Item_Weight            4705 non-null  float64
2   Item_Fat_Content       5681 non-null  object

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3   Item_Visibility      5681 non-null   float64
4   Item_Type            5681 non-null   object
5   Item_MRP             5681 non-null   float64
6   Outlet_Identifier     5681 non-null   object
7   Outlet_Establishment_Year 5681 non-null   int64
8   Outlet_Size          4075 non-null   object
9   Outlet_Location_Type  5681 non-null   object
10  Outlet_Type          5681 non-null   object
dtypes: float64(3), int64(1), object(7)
memory usage: 488.3+ KB

```

```
[7]: # Descriptive Statistics of the Dataset
df.describe()
```

```

[7]:      Item_Weight  Item_Visibility  Item_MRP  Outlet_Establishment_Year
count  4705.000000      5681.000000  5681.000000      5681.000000
mean     12.695633         0.065684   141.023273      1997.828903
std       4.664849         0.051252    61.809091         8.372256
min       4.555000         0.000000    31.990000      1985.000000
25%       8.645000         0.027047    94.412000      1987.000000
50%      12.500000         0.054154   141.415400      1999.000000
75%      16.700000         0.093463   186.026600      2004.000000
max      21.350000         0.323637   266.588400      2009.000000

```

```
[8]: # check for duplicates
duplicates = df.duplicated()
```

```

[9]: # checking for null values
null_counts= df.isnull().sum()
if null_counts.sum()>0:
    print(null_counts[null_counts>0])

```

```

Item_Weight      976
Outlet_Size      1606
dtype: int64

```

```
[10]: 976/5681
```

```
[10]: 0.17180073930646014
```

```
[11]: 1606/5681
```

```
[11]: 0.28269670832599897
```

```

[14]: #If the number of entries is N, you have upto 50% missing values ---we can
      ↪replace those rows with mean, median, mode
df['Item_Weight'] = df['Item_Weight'].fillna(df['Item_Weight'].median())

```

```
#Fill missing values in 'Outlet_Size' coloumn with mode
df['Outlet_Size'] = df['Outlet_Size'].fillna(df['Outlet_Size'].mode()[0])
```

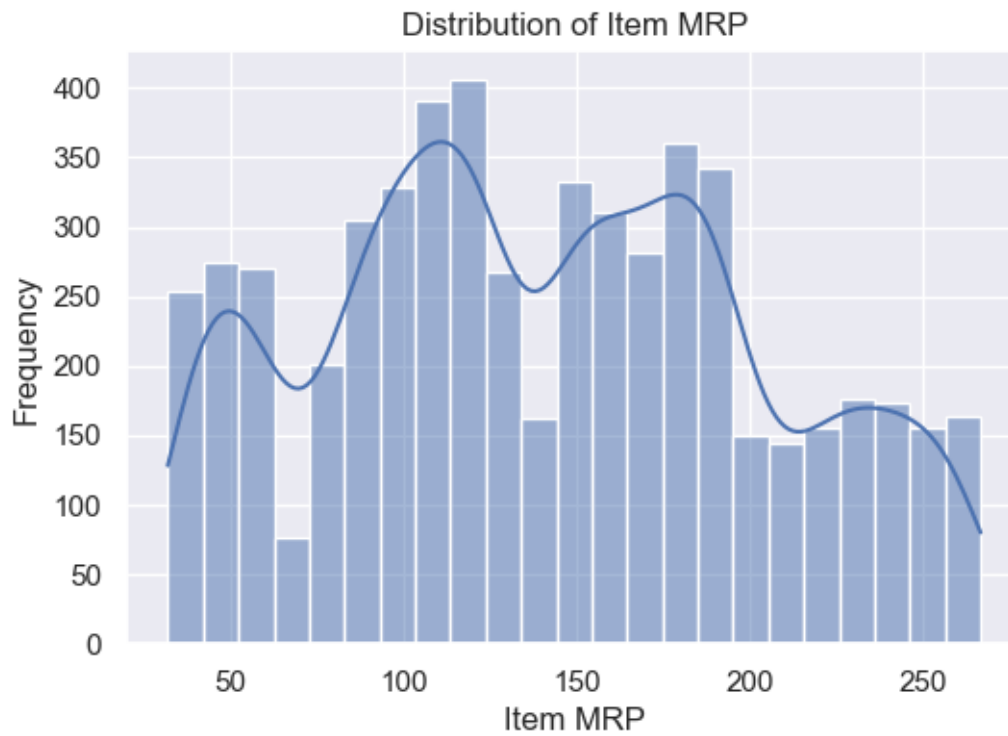
```
[13]: # check for missing values again after filling
print(df.isnull().sum())
```

```
Item_Identifier      0
Item_Weight          0
Item_Fat_Content     0
Item_Visibility      0
Item_Type            0
Item_MRP             0
Outlet_Identifier    0
Outlet_Establishment_Year  0
Outlet_Size          0
Outlet_Location_Type 0
Outlet_Type          0
dtype: int64
```

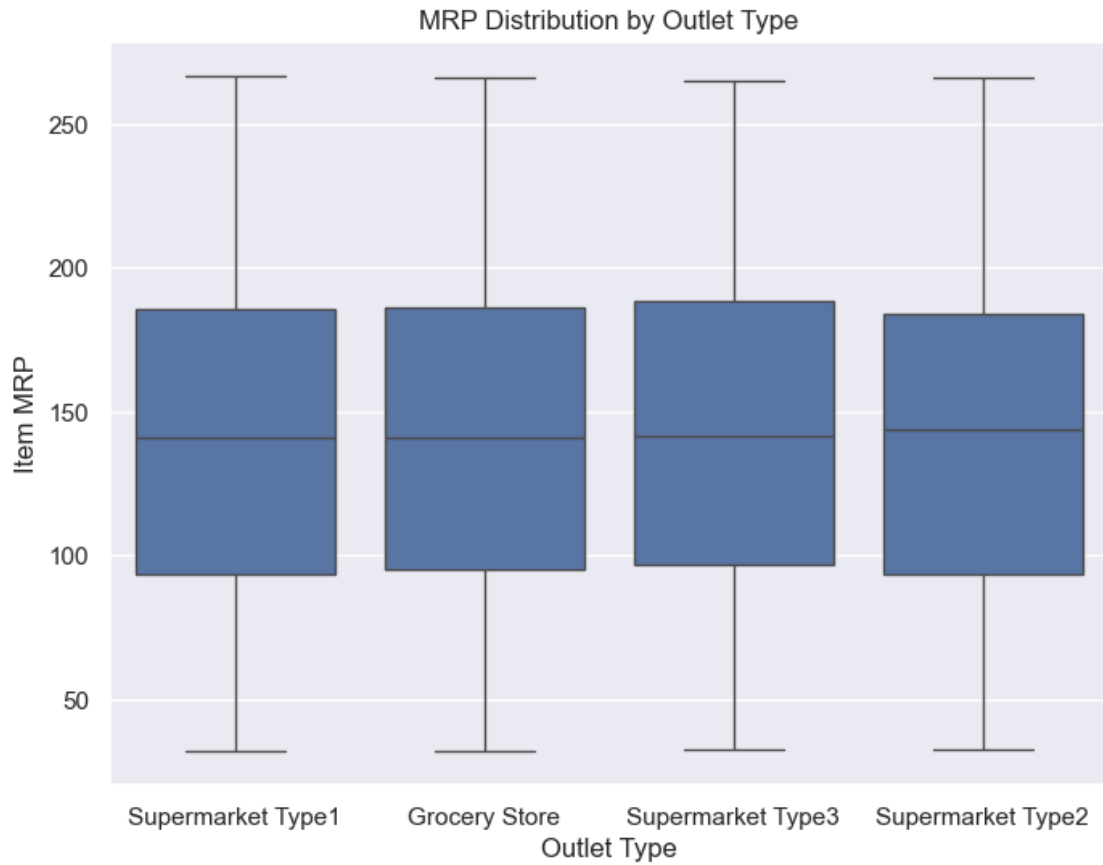
```
[16]: print(df.columns)
```

```
Index(['Item_Identifier', 'Item_Weight', 'Item_Fat_Content', 'Item_Visibility',
      'Item_Type', 'Item_MRP', 'Outlet_Identifier',
      'Outlet_Establishment_Year', 'Outlet_Size', 'Outlet_Location_Type',
      'Outlet_Type'],
      dtype='object')
```

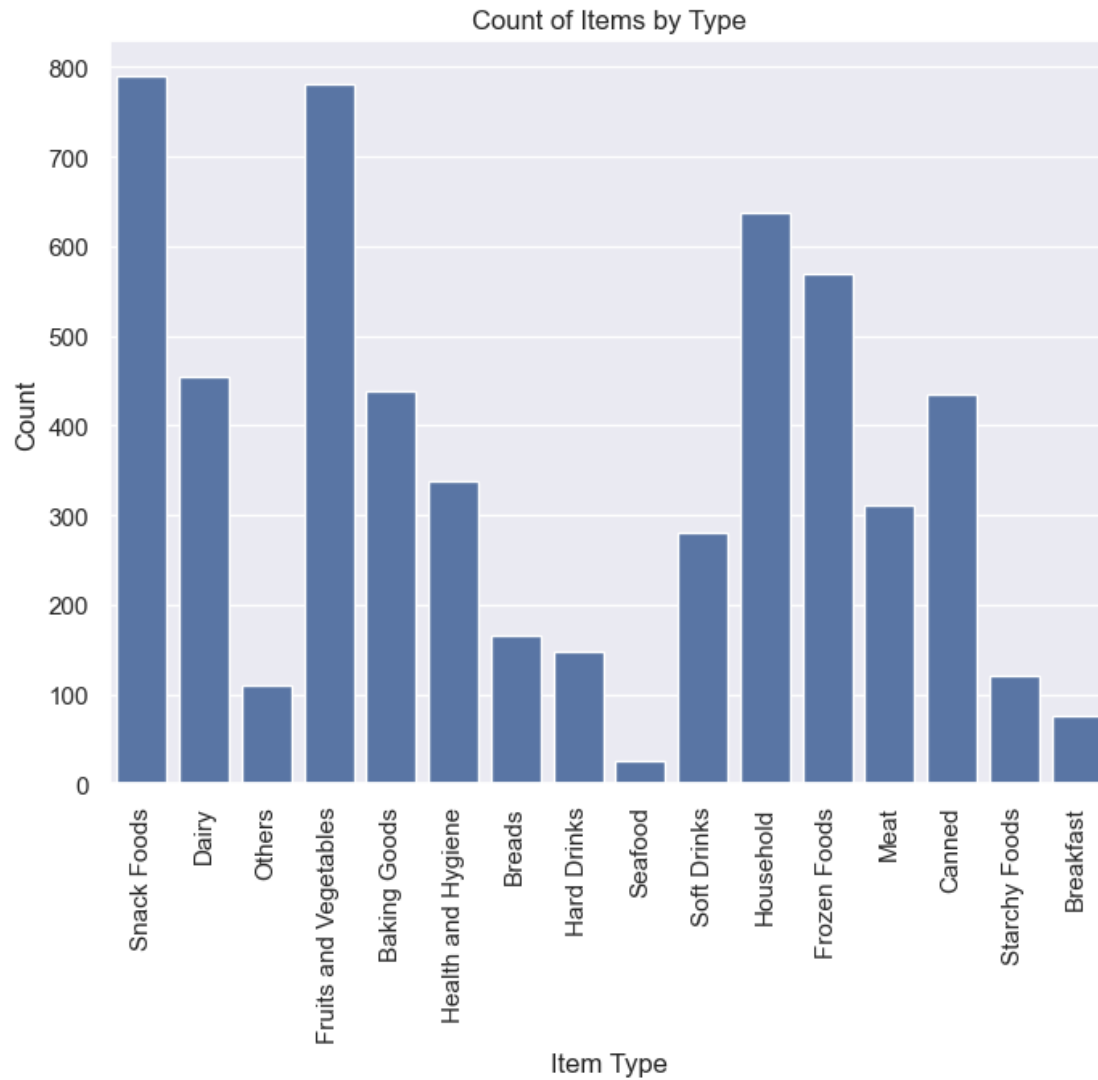
```
[18]: # Histogram
plt.figure(figsize=(6,4))
sns.histplot(df['Item_MRP'], kde=True)
plt.title('Distribution of Item MRP')
plt.xlabel('Item MRP')
plt.ylabel('Frequency')
plt.show()
```



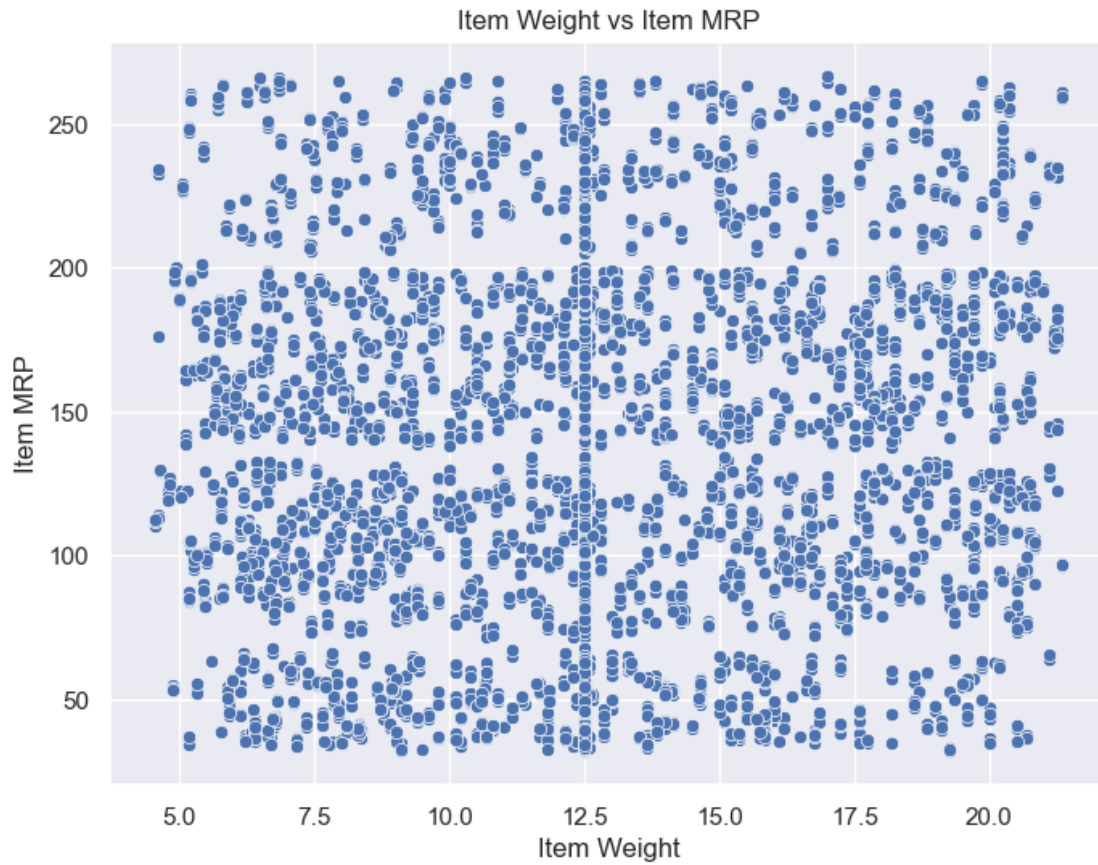
```
[19]: # Box plot
plt.figure(figsize=(8, 6))
sns.boxplot(x='Outlet_Type', y='Item_MRP', data=df)
plt.title('MRP Distribution by Outlet Type')
plt.xlabel('Outlet Type')
plt.ylabel('Item MRP')
plt.show()
```



```
[20]: # Bar plot
plt.figure(figsize=(8,6))
sns.countplot(x='Item_Type', data=df)
plt.title('Count of Items by Type')
plt.xlabel('Item Type')
plt.ylabel('Count')
plt.xticks(rotation=90)
plt.show()
```



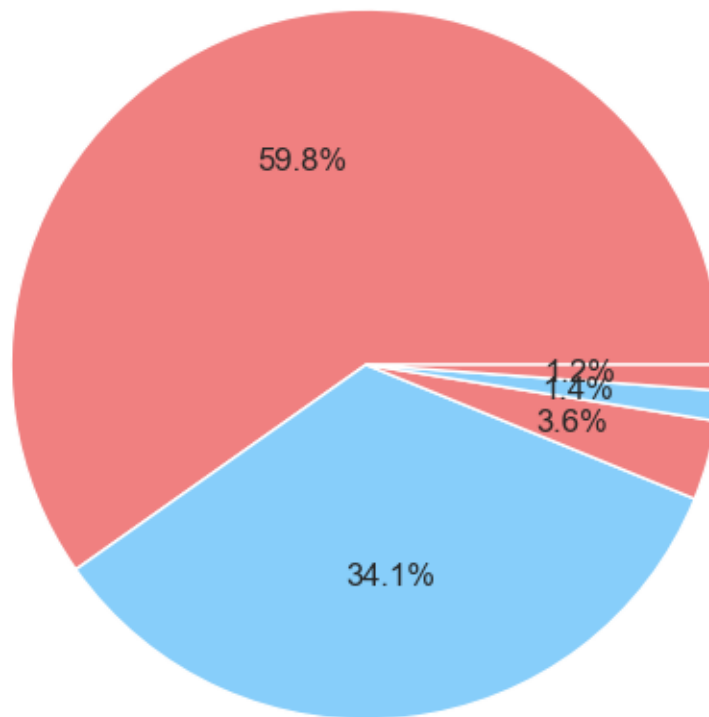
```
[21]: # Scatter plot
plt.figure(figsize=(8, 6))
sns.scatterplot(x='Item_Weight', y='Item_MRP', data=df)
plt.title('Item Weight vs Item MRP')
plt.xlabel('Item Weight')
plt.ylabel('Item MRP')
plt.show()
```



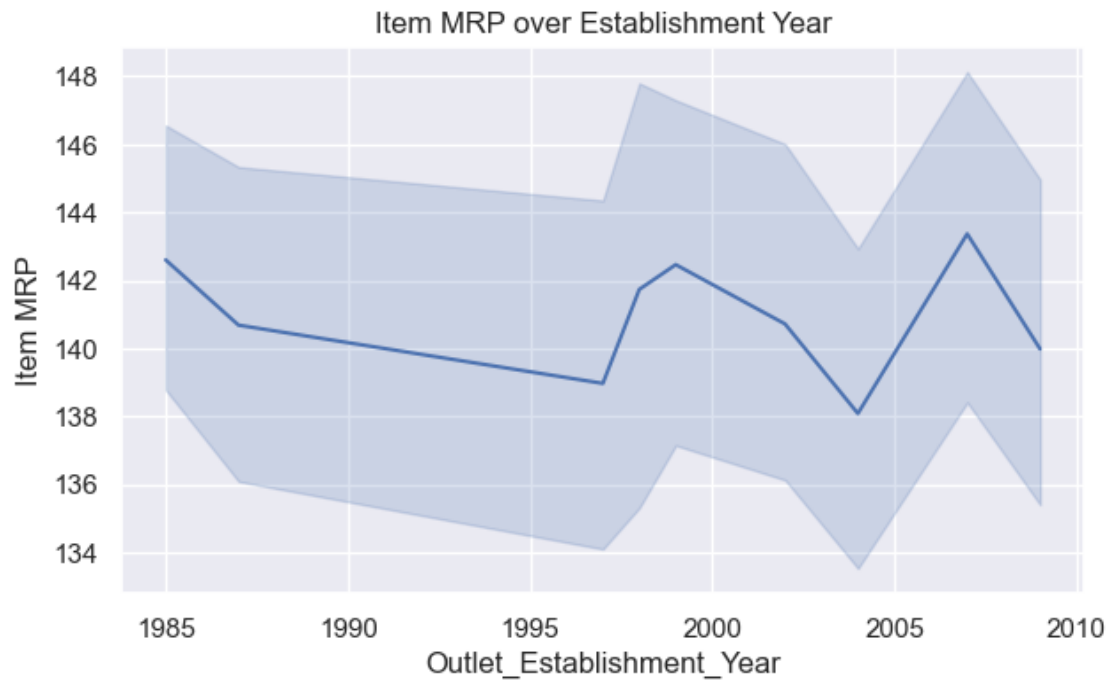
```
[40]: # pie plot
plt.figure(figsize=(6, 6))
plt.pie(df['Item_Fat_Content'].value_counts(), autopct='%1.1f%%',
        colors=['lightcoral', 'lightskyblue'])
plt.title('Proportion of Item Fat Content')
plt.ylabel('')

plt.show()
```

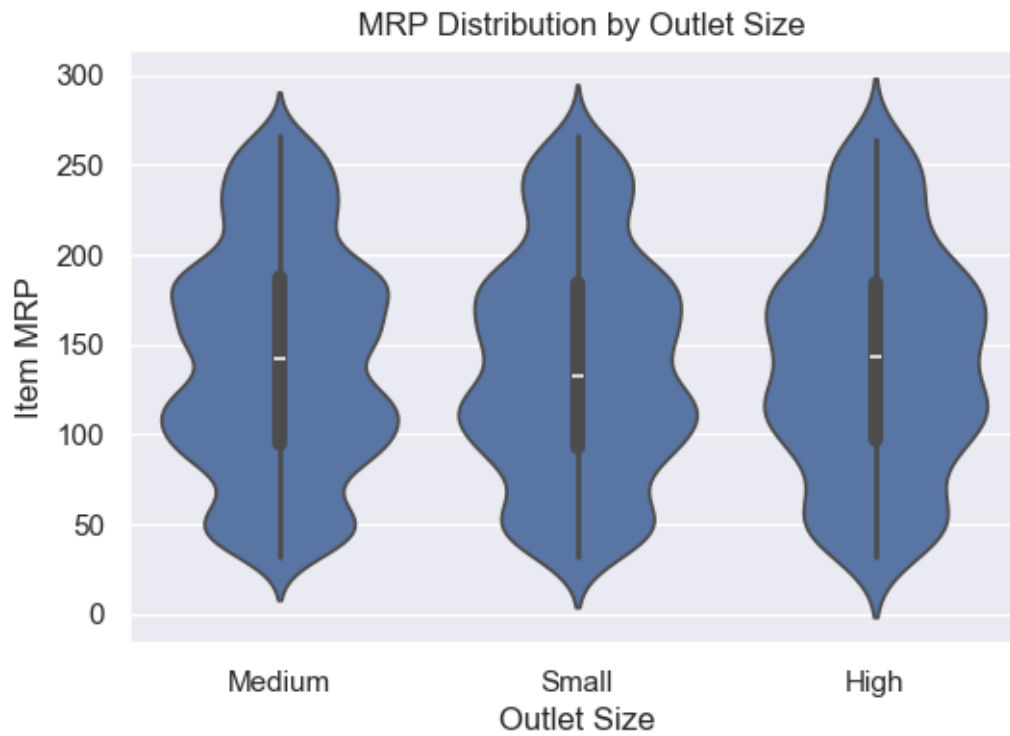
Proportion of Item Fat Content



```
[41]: # line plot
plt.figure(figsize=(7, 4))
sns.lineplot(x='Outlet_Establishment_Year', y='Item_MRP', data=df)
plt.title('Item MRP over Establishment Year')
plt.xlabel('Outlet_Establishment_Year')
plt.ylabel('Item MRP')
plt.show()
```

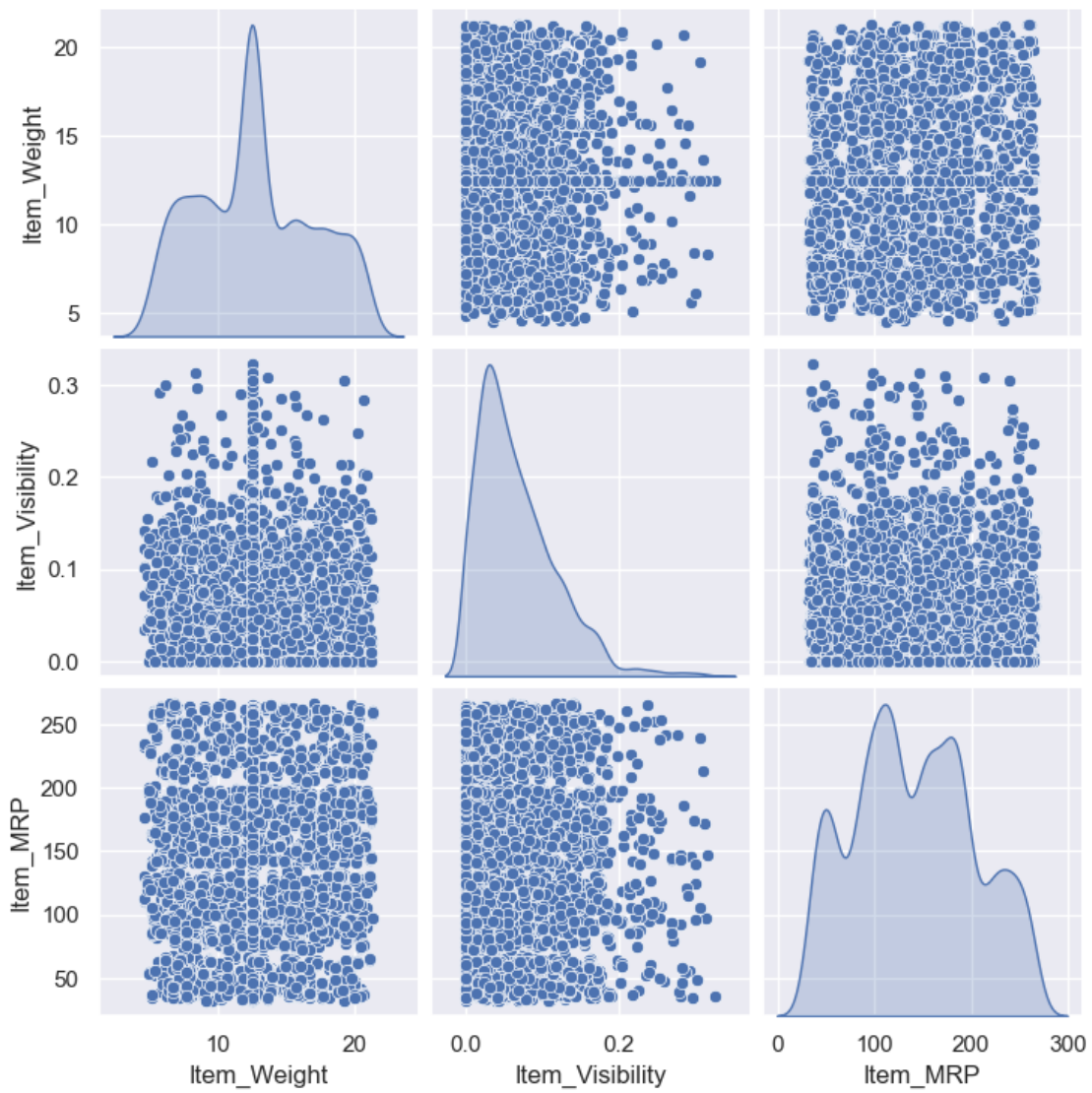


```
[43]: # violin plot
plt.figure(figsize=(6, 4))
sns.violinplot(x='Outlet_Size', y='Item_MRP', data=df)
plt.title(' MRP Distribution by Outlet Size')
plt.xlabel('Outlet Size')
plt.ylabel('Item MRP')
plt.show()
```



```
[44]: # pair plot
sns.pairplot(df[['Item_Weight', 'Item_Visibility', 'Item_MRP']],
             diag_kind='kde')
plt.suptitle('Pairwise Relationship between Numerical Variables', y=1.02)
plt.show()
```

Pairwise Relationship between Numerical Variables



[]: